

Real Analysis General Exam January 2022

Instructions. 2 hours. Closed book examination. Be neat in your presentation. When invoking a theorem from previous courses, name the theorem and thoroughly check its hypotheses. You must solve a significant portion of each of the three problems in order to pass the exam.

Problem 1

Let $(X, \mathcal{M}, \mu)$ be a measure space. Let (f_n) be a sequence of nonnegative functions in $L^1(X, \mathcal{M}, \mu)$ and let f be a nonnegative function in $L^1(X, \mathcal{M}, \mu)$. Suppose that

$$\int_X f_n \, d\mu \rightarrow \int_X f \, d\mu$$

and that $f_n \rightarrow f$ pointwise. Prove that f_n converges to f in $L^1(X, \mathcal{M}, \mu)$. Hint: consider $g_n = \min(f, f_n)$.

Problem 2

Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $p \in [1, \infty)$.

(a) Let (f_n) be a sequence of functions in $L^p(X, \mathcal{M}, \mu)$ and let f be a function in $L^p(X, \mathcal{M}, \mu)$.

Suppose that f_n converges to f in $L^p(X, \mathcal{M}, \mu)$. Prove that there exists a subsequence (f_{n_k}) such that for μ -almost all x , $\lim_{k \rightarrow \infty} f_{n_k}(x) = f(x)$. Hint: remember the proof of completeness of L^p .

(b) Let h be a measurable function on X . Let

$$D = \{f \in L^p(X, \mathcal{M}, \mu) \mid hf \in L^p(X, \mathcal{M}, \mu)\}$$

Let (f_n) be a sequence of elements of D , and let $f, g \in L^p(X, \mathcal{M}, \mu)$ be such that f_n converges to f in L^p , and hf_n converges to g in L^p . Show that $f \in D$ and $g = hf$.

Problem 3

For μ a Borel probability measure on $\mathbb{R}$, we will denote by $\hat{\mu}$ the function $\mathbb{R} \rightarrow \mathbb{C}$ given by

$$\hat{\mu}(t) = \int_{\mathbb{R}} e^{itx} \, d\mu(x)$$

We will also adopt the notational convention $\text{sinc}(x) = \frac{\sin x}{x}$ if $x \neq 0$ and $\text{sinc}(0) = 1$.

(a) Show that $\widehat{\mu}$ is a bounded continuous function.

(b) Let $\delta > 0$. Show that

$$\frac{1}{2\delta} \int_{-\delta}^{\delta} (1 - \text{Re}(\widehat{\mu}(t))) dt = \int_{\mathbb{R}} (1 - \text{sinc}(\delta x)) d\mu(x)$$

(c) Show that for all $u \in \mathbb{R}$,

$$1 - \text{sinc}(u) \geq \frac{1}{2} \chi_{(-\infty, -2) \cup (2, \infty)}(u)$$

and deduce that

$$\mu\left(\{x \in \mathbb{R} \mid |x| > 2\delta^{-1}\}\right) \leq \frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \text{Re}(\widehat{\mu}(t))) dt$$

(d) Let μ_n be a sequence of Borel probability measures on $\mathbb{R}$. Suppose that for all t , the limit $\Phi(t) = \lim_{n \rightarrow \infty} \widehat{\mu_n}(t)$ exists and that the resulting function $\Phi(t)$ is continuous at $t = 0$. Prove that for all $\epsilon > 0$, there exists a compact set K inside $\mathbb{R}$ such that, for all n , $\mu_n(K) \geq 1 - \epsilon$.